Estimating Winter Honey Bee Colony Losses*

Design and simulation of a stratified sample survey of registered beekeepers

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Abstract

National surveys of honey bee colony losses rely mostly on self-selected samples, whose accuracy cannot be quantified. This study designs and evaluates a probability survey of total colony losses over winter 2025/26 by the 10,000 registered beekeepers of a hypothetical country, with a 95% confidence interval (CI) whose half-width is at most 5% of a baseline value. All data are simulated, calibrated to the COLOSS survey of winter 2019/20. Beekeepers are stratified by operation size with Neyman allocation, the 320 commercial operations are taken in full, and the total is estimated with a ratio estimator using the register's colony counts. Planned for a 77% response rate, with the expansion estimator's larger variance as a safety margin, the survey needs 1,887 respondents from 2,452 issued beekeepers, whereas simple random sampling without replacement (SRSWOR) would need 7,289 respondents even with full response. One simulated survey estimates that 40,090 colonies were lost, 14.8% of colonies going into winter, against a true total of 41,853. Over 10,000 repeated samples the estimator is essentially unbiased, meets the 5% bound in 97% of samples and covers the true total in 94.8%, against 91.4% for SRSWOR. Without the colony counts, the expansion estimator meets the bound in only 27% of samples, as the prior understated commercial variability. The commercial response rate is the main risk: below 63% on the planning values, no sample size reaches the target. Stratification by operation size with a take-all commercial stratum and ratio estimation is an efficient basis for national colony-loss surveys.

Keywords: stratified random sampling; Neyman allocation; take-all stratum; ratio estimation; non-response weighting; Monte Carlo simulation; honey bee colony losses

1 Introduction and Motivation

1.1 Background (A2)

Honey bees produce honey and play a critical role as pollinators of crops and natural vegetation (Pirk et al., 2014). The COLOSS (Prevention of honey bee COlony LOSSes) network runs international monitoring of colony losses with a standardised questionnaire, so that results can be compared across countries (Gray et al., 2023; van der Zee et al., 2013). The 30,491 beekeepers outside New Zealand in the COLOSS survey of winter 2019/20 lost 18.1% of their colonies, with national loss rates from 7.4% to 36.5% (Gray et al., 2023). A reliable national total shows how much honey production and pollination capacity is lost each winter, and helps beekeepers' associations and veterinary services to target support and disease control.

1.2 Statement of the problem

This project designs a sample survey to estimate how many honey bee colonies the registered beekeepers of a hypothetical country lost over winter 2025/26, from 1 October 2025 to 31 March 2026. The country takes part in the COLOSS monitoring and uses its case definitions and questionnaire. Unlike the self-selected samples common in colony-loss monitoring (Section 2.1), however, the survey must be

*Code, data and README: https://github.com/KingBifteki/honey-bee-colony-loss-survey

a probability sample: its design must differ from simple random sampling, its 95% confidence interval (CI) must have a half-width of at most 5% of a baseline value (Section 3.2.1), and simple random sampling without replacement (SRSWOR) serves as the benchmark.

1.3 Aim and objectives (A1)

The aim is to design, draw and analyse a stratified sample that meets this requirement, and to evaluate the design by simulation, through four questions:

- How many colonies did registered beekeepers lose in total over winter 2025/26, and what share of the colonies going into winter was lost?
- How are the losses spread across hobby, side-line and commercial operations?
- How many beekeepers must be sampled, and how should they be allocated across strata, to meet the precision requirement once non-response is allowed for?
- Does stratification by operation size with Neyman allocation reach the required precision with far fewer respondents than SRSWOR, and are its CIs valid in repeated sampling?

Section 2 reviews the literature and sets out the methods, Section 3 presents the results, and Section 4 draws conclusions and makes recommendations.

2 Review of Methods

2.1 Literature review

2.1.1 Colony-loss surveys

Colony-loss surveys agree on how losses are measured but differ in the period they cover and in how they group operations. The COLOSS network reports the loss rate as the share of colonies going into winter that were lost, with confidence intervals from quasi-binomial models that allow for extra-binomial variation between operations (Gray et al., 2020, 2023; van der Zee et al., 2013). Its questionnaire covers the northern-hemisphere winter, so Pirk et al. (2014) modified it for the southern hemisphere, and Nganso et al. (2025) applied it to the active beekeeping season in nine Sub-Saharan African countries. The United States survey of vanEngelsdorp et al. (2008) grouped operations into 1–50, 51–499 and 500 or more colonies, against the COLOSS classes of up to 50, 51–150 and more than 150, so loss rates by operation size cannot be compared directly across surveys.

The findings agree on operation size but not on migration: in the COLOSS surveys, operations with more than 150 colonies in winter 2018/19, and with more than 50 colonies in winter 2019/20, had significantly lower loss rates than smaller ones, a difference both reports link to management (Gray et al., 2020, 2023). The United States survey, by contrast, found no difference in loss by operation size (vanEngelsdorp et al., 2008). Beekeepers who migrated their colonies lost more than those who did not in winter 2018/19 (Gray et al., 2020) and in South Africa (Pirk et al., 2014), but less in winter 2019/20 (Gray et al., 2023), and Gray et al. (2020) note that the migration effect varies from year to year and between countries.

The surveys rely largely on non-probability samples, with each COLOSS country organising its own means of contacting beekeepers, many now mostly or only through online surveys (Gray et al., 2023). The South African survey was a survey of convenience with snowball recruitment, answered by about 4% of registered beekeepers (Pirk et al., 2014); in the United States, apiary inspectors telephoned beekeepers they felt were representative of their state's industry (vanEngelsdorp et al., 2008); and the respondents in Africa made up less than 1% of the beekeepers in the nine countries (Nganso et al., 2025). Van der Zee et al. (2013) urge a shift from self-selected samples towards at least simple random sampling, as a random sample gives results whose accuracy can be quantified, and report response rates from 7.5%, for an open invitation in a national beekeeping journal, to 77% where beekeepers were contacted personally.

2.1.2 Sampling methods

Neyman (1934) showed that, for a fixed total sample size, the variance of the stratified estimator is smallest when the sample in each stratum is proportional to the stratum size times the stratum SD, $N_h S_h$, and that the SDs can be estimated beforehand from a preliminary enquiry. Neyman allocation is commonly used when the cost per unit is constant and the stratum variances differ substantially, as in skewed populations, which can be stratified with a certainty (take-all) stratum of the largest units, in which every unit is examined (Gunning et al., 2008). For non-response, most surveys inflate the weights within weighting classes by the inverse of the class response rate (Brick, 2013). This removes the bias if every unit in a class has the same probability of responding, the missing-at-random (MAR) assumption (Brick, 2013; Kalton & Flores-Cervantes, 2003), and works best when the classes are related to both response and the survey variables (Kalton, 1983, as cited in Brick, 2013). Brick (2013) also notes that response rates may not predict non-response bias; in this study the response rate matters mainly for precision in the take-all stratum (Section 3.2.2).

2.2 Target population and parameter (A2–A4)

For beekeeper i, let m_i be the number of colonies going into winter on 1 October 2025 and y_i the number of those colonies lost by 31 March 2026, the COLOSS winter period (Nganso et al., 2025). Following the COLOSS case definition (van der Zee et al., 2013), a colony counts as lost when:

- it no longer has any living bees, for example because it died or absconded;
- its population became too small for normal activities such as brood rearing and foraging; or
- it had queen problems that could not be solved, such as a drone-laying queen.

The definition also counts colonies lost to theft, fire, flooding or similar causes unrelated to health. These losses are excluded here, as in the COLOSS analysis of operation size, because they are largely beyond the beekeeper's control (Gray et al., 2023). The primary parameter is the total number of colonies lost,

$$T = \sum_{i=1}^{N} y_i, \qquad N = 10{,}000.$$

The target population is the 10,000 beekeepers on the national register who had at least one colony going into winter on 1 October 2025. Unregistered beekeepers fall outside it, so T refers to registered beekeepers only. They are described by two institutional characteristics, registration and operation-size class (hobby, side-line or commercial), by their number of colonies and by migratory practice, a management practice with a geographic side, as migrating beekeepers move their colonies between sites; region is not modelled (Section 2.4). Van der Zee et al. (2013) name registers held through beekeeping associations, census registers and veterinary services as possible sampling frames for colony-loss surveys, while noting that few countries have a representative list. The register here is assumed to record each beekeeper's contact details and number of colonies as at 1 October 2025, so $M = \sum_i m_i$ is known and the loss rate T/M follows from T. Personal characteristics such as age or beekeeping experience are assumed not to be recorded, so they cannot serve as stratifiers; Table 1 separates the population concepts that follow from these choices.

Table 1: Key population concepts.

Concept	Definition
Target population	The 10,000 registered beekeepers, about whom conclusions are drawn.
Sampling frame	The national register of beekeepers, with contact details and number of colonies.
Sampled population	Eligible registered beekeepers who can be contacted (beekeepers with out-of-date contact details are missed).
Sampling unit	The registered beekeeper (one beekeeping operation); element sampling, no clusters.
Observation unit	The beekeeper's whole operation, for which colonies and losses are reported, as in the COLOSS survey (Gray et al., 2023).

2.3 Notation

Strata $h = 1, 2, 3$ are hobby operations (HOB, up to 50 colonies), side-line operations (SID, 51 to 150) and commercial operations (COM, more than 150), the size classes of Gray et al. (2023) with the names used by van der Zee et al. (2013). N_h is the number of beekeepers in stratum h ($\sum_h N_h = N$), $W_h = N_h/N$ its stratum weight and M_h its number of colonies. $\bar{Y}_h$, S_h and S_h^2 are the stratum mean, standard deviation (SD) and variance of y_i, and $T_h = N_h\bar{Y}_h$ is the stratum total; $\bar{Y} = T/N$ and S^2 are the population mean and variance. n_h is the planned number of respondents in stratum h (Section 3.2), $n = \sum_h n_h$, n_h^* the number of beekeepers issued and r_h the number who responded. Lowercase letters ($\bar{y}_h$, s_h^2) denote sample values, and $\bar{y}_{st} = \sum_h W_h\bar{y}_h$ is the stratified sample mean.

2.4 Simulated finite population (A5)

The country is hypothetical, so its population is calibrated to the COLOSS survey of winter 2019/20 (Gray et al., 2023): the split of beekeepers and colonies across operation sizes, the loss rate of each size class and the share of beekeepers who migrate all come from that survey (Appendix A). Region is not modelled, as the calibration uses COLOSS results pooled over 36 countries.

Each beekeeper's number of colonies m_i is drawn from a distribution over their size class with mean 10.8, 88.2 or 371.8 colonies, the class means computed from the numbers of colonies and beekeepers reported by Gray et al. (2023). That survey gives no distribution within the size classes, so the shapes are assumptions: the numbers of hobby and side-line operations fall geometrically with size, and commercial operations follow a long-tailed lognormal distribution, capped at 5,000 colonies, that allows for a few very large operations. Each beekeeper migrates with probability 0.183, since 81.7% of COLOSS beekeepers did not migrate their colonies (Gray et al., 2023; Appendix A). That survey does not report migration by operation size, so the two are assumed unrelated.

Each beekeeper's loss risk p_i follows a Beta distribution whose mean is the COLOSS loss rate of their size class, 16.6%, 14.6% or 15.2% (Gray et al., 2023), with SD 0.15 (an assumption), and $y_i \sim \text{Binomial}(m_i, p_i)$, all drawn in `R/01_population.R` with `set.seed(4102)`. This beta-binomial model suits loss counts: each colony either survives the winter or is lost, and losses vary more between operations than a binomial model allows, which is why COLOSS analyses use quasi-binomial models (Section 2.1.1). Migration is left out of the loss model: its effect was significant in winter 2019/20 (15.2% for migrants against 16.1%, $p < 0.001$; Gray et al., 2023), but its direction changes between years and countries (Section 2.1.1).

2.5 Choice of design (A6)

The primary design is stratified random sampling by operation size, with an independent SRSWOR in each stratum and Neyman allocation, $n_h \propto N_h S_h$. Neyman allocation can ask for more beekeepers than a stratum contains: with the planning values it would assign 457 respondents to the commercial stratum, which has only 320 beekeepers (Section 3.2). That stratum is therefore taken in full (a take-all stratum), and the rest of the sample is shared among the other strata; every stratum keeps at least 30 respondents. Van der Zee et al. (2013) recommend stratified sampling for colony-loss surveys, name operation size as one basis for the strata so that hobby and commercial operations are both covered, and report Neyman allocation in the Scottish surveys of 2008 (in a modified form) and 2011.

- Population structure: losses per beekeeper differ sharply between operation-size groups although the loss rates are similar (Section 3.1), as colony counts are extremely skewed, a few large operations holding a large share of the colonies and losses (Figure 1). Stratifying by operation size separates these groups and puts the few large operations in a stratum of their own. The boundaries are the COLOSS size classes, the only ones with published loss rates. Splitting a class on colony counts would improve efficiency further: under the planning model, a split of the hobby class at 12 colonies cuts the respondents needed from 1,255 to 975, and such boundaries can be optimised with the Lavallée and Hidiroglou algorithm (Gunning et al., 2008). The design keeps the COLOSS classes and uses the colony counts, which drive the differences in losses, in a ratio estimator instead (Section 2.9).

- Efficiency and precision: the estimated total $\hat{T} = N\bar{y}_{st}$ has variance $N^2 V(\bar{y}_{st})$, which Neyman allocation minimises (Section 2.1.2) by sampling most where $N_h S_h$ is largest. Under full response the take-all stratum then contributes no variance; non-response brings part of it back.
- Cost: operation size is already on the register for every beekeeper, so stratifying needs no extra data collection. Every stratum uses the same questionnaire and interview modes, so the cost per interview c_h is assumed equal across strata, and the cost-optimal allocation $n_h \propto N_h S_h / \sqrt{c_h}$ then reduces to Neyman allocation.
- Operational convenience: selection is a separate SRSWOR within each stratum of the size-sorted register, done in one step with standard software (Section 2.6).
- Alternatives: cluster sampling would raise the variance, systematic sampling gives only implicit stratification, and selection with probability proportional to colony counts would take every commercial operation with certainty anyway, as the take-all stratum does.

The design is not self-weighting, so the analysis must use the design weights defined in Section 2.6.

2.6 Selection and data collection (A7)

The register is cleaned (duplicates and beekeepers with no colonies removed) and sorted by stratum. The planned n_h (Section 3.2) are inflated for the anticipated response rate to give the issued n_h^*. An SRSWOR of n_h^* beekeepers is drawn in each stratum with `sampling::strata()`, so each beekeeper has inclusion probability $\pi_{hi} = n_h^*/N_h$ and design weight $d_{hi} = 1/\pi_{hi}$. After fieldwork the weights are adjusted within each stratum to $w_{hi} = N_h/r_h$, the weighting-class adjustment of Section 2.1.2 with the strata as classes.

The survey would use the COLOSS standardised questionnaire (van der Zee et al., 2013), which records colonies going into winter, those lost by cause and whether colonies were migrated (Gray et al., 2023). The questionnaire would be pretested, and selected beekeepers then invited by e-mail and SMS to complete it online, with two reminders and computer-assisted telephone interviewing (CATI) for those who do not reply. Fieldwork would run in April and May 2026, straight after the winter, in line with the annual COLOSS spring survey (Gray et al., 2023), with consent under the country's data-protection law; in this project it is simulated (Section 3.3).

2.7 Sources of non-sampling error (A8)

The main sources of non-sampling error are coverage error (duplicate entries, registered beekeepers who no longer keep bees, outdated contact details or colony counts), non-response, response error (recall error, ambiguous cases such as merged or split colonies) and processing error (data-entry errors); Table B1 in Appendix B gives their likely effects on $\hat{T}$ and their mitigation. The design-based SEs of Sections 3.4 and 3.5 do not measure them. The simulation assumes that the register covers the target population exactly and that non-response is random within strata; non-response related to the losses themselves would leave bias after weighting (Section 3.5.2).

2.8 Sample-size formula (B1)

With an SRSWOR of n_h respondents in each stratum, $\hat{T} = \sum_h N_h \bar{y}_h$ has variance

$$V(\hat{T}) = \sum_h N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{S_h^2}{n_h}.$$

The 95% CI has half-width $z\sqrt{V(\hat{T})}$ with $z = 1.96$, so a half-width of at most $e\,T_0$, where $e = 0.05$ and T_0 is the baseline value, requires $V(\hat{T}) \le V_0 = (e\,T_0/z)^2$. Under Neyman allocation, $n_h = n\,N_h S_h / \sum_k N_k S_k$, the smallest total sample size

$$n = \frac{\left(\sum_h N_h S_h\right)^2}{V_0 + \sum_h N_h S_h^2},$$

where the term $\sum_h N_h S_h^2$ comes from the finite population correction. With a take-all stratum (Section 2.5), the formula is applied to the remaining strata. For the SRSWOR benchmark, $n =$

$n_0/(1+n_0/N)$ with $n_0 = (zNS/(eT_0))^2$, where S is the SD of colonies lost across all beekeepers. For the primary estimator, the ratio estimator of Section 2.9, the same formulas apply approximately with S_h replaced by the smaller SD S_{eh} of the residuals about the colony counts; the plan uses S_h, which gives it a safety margin.

2.9 Estimation and simulation design (D1, E1, E2)

The respondents are analysed with the R `survey` package (Lumley, 2004), using a design object with the three strata, no clusters, the adjusted weights $w_{hi} = N_h/r_h$ and the finite population correction from N_h; the code is in Appendix C. The respondents in each stratum are thus treated as an SRSWOR of r_h from N_h, which holds when non-response is random within strata (Section 3.2.2).

The register also gives every beekeeper's colonies m_i and the stratum totals M_h. The primary estimator, the separate ratio estimator $\hat{T}_R = \sum_h M_h \bar{y}_h/\bar{m}_h$ with $\bar{m}_h$ the respondents' mean number of colonies, uses them and is computed with `svyratio()`. Its approximate variance has the form of $V(\hat{T})$ with S_h^2 replaced by the variance of the residuals $y_i - R_h m_i$, where $R_h = T_h/M_h$, so it gains when losses rise roughly in proportion to colony counts. Planning the sample size with the larger variance of the expansion estimator $\hat{T}$ gives $\hat{T}_R$ a safety margin (Section 3.3.1), and $\hat{T}$ is reported alongside it for comparison.

`R/05_simulation.R` repeats the complete procedure 10,000 times on the fixed population, with `set.seed(1021)`. Each repetition draws a new stratified sample of 1,565, 567 and 320 beekeepers, simulates 77% response within strata, and computes $\hat{T}$ and $\hat{T}_R$, their estimated SEs and their 95% CI limits with the closed forms that `svytotal()` and `svyratio()` use (checked against them in `R/04_estimation.R`). Each repetition also draws an SRSWOR of 2,452 beekeepers with the same response process. The estimates, SEs and CI limits from all 20,000 samples are saved in `output/tables/sim_reps.csv`. Following Morris et al. (2019), each estimator is assessed by its bias, empirical SE, average estimated SE, root mean squared error (RMSE), CI coverage and average CI width, with Monte Carlo SEs for the bias and the coverage; the Monte Carlo SE of an estimated coverage $\hat{p}$ is $\sqrt{\hat{p}(1-\hat{p})/R}$ for R repetitions.

3 Analysis and Results

3.1 Data description (A5)

The population lost $T = 41{,}853$ colonies out of $M = 270{,}987$, a total loss of 15.4%, close to the COLOSS loss rate of 15.6% excluding natural disasters (11.1% dead or empty plus 4.5% with queen problems; Gray et al., 2023). The planner knows neither these values nor those in Table 2, and instead uses the COLOSS loss rates by size class and the register's colony counts (Section 3.2.1).

Table 2: True stratum parameters, unknown to the planner. $\bar{Y}_h$, S_h, S_h^2 and T_h refer to y_i, colonies lost per beekeeper; the loss rate is T_h/M_h, with the COLOSS rate of each class from Gray et al. (2023).

								Loss rate	
Stratum	N_h	W_h	M_h	$\bar{Y}_h$	S_h	S_h^2	T_h	Sim.	COLOSS
HOB	9,020	0.902	97,216	1.79	2.86	8.2	16,115	16.6%	16.6%
SID	660	0.066	57,062	12.08	13.72	188.2	7,974	14.0%	14.6%
COM	320	0.032	116,709	55.51	87.13	7,592.0	17,764	15.2%	15.2%
Population	**10,000**	**1.000**	**270,987**	**4.19**	**18.86**	**355.6**	**41,853**	**15.4%**	**15.6%**

The strata differ sharply (Table 2, Figure 1): commercial operations are only 3.2% of beekeepers but hold 43% of the colonies and 42% of the colonies lost. Mean losses are 1.79, 12.08 and 55.51 colonies per beekeeper, and the between-stratum share of the variance is $\eta^2 = 1 - \sum_h (N_h - 1)S_h^2/[(N-1)S^2] = 26.3\%$. Within strata, hobby operations are far more homogeneous than the population as a whole (SD 2.86 colonies, against 18.86 overall) and side-line operations somewhat more so (SD 13.72). Commercial

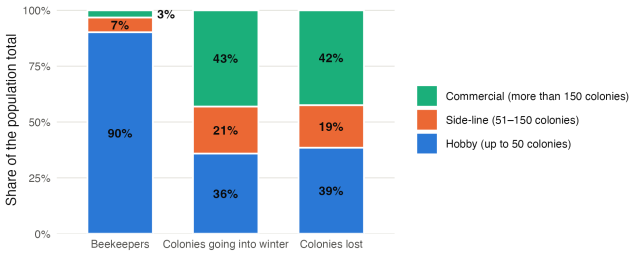


Figure 1: Share of beekeepers, colonies and colonies lost by stratum. Commercial beekeepers manage 365 colonies each on average, against 11 for hobby beekeepers.

operations remain highly variable (SD 87.13) and hold 92% of the within-stratum variation, which adds sampling variance only through non-response, as the stratum is taken in full (Section 3.2.2).

3.2 Sample size and allocation (B2–B10)

3.2.1 Baseline and minimum sample size (B2, B3)

The planner knows each beekeeper's colonies m_i from the register and the COLOSS loss rate p_h of each size class (Gray et al., 2023), so the baseline is $T_0 = \sum_h p_h M_h = 42{,}209$ colonies and the target half-width of the 95% CI is 2,110 colonies. The true total, unknown to the planner, is 41,853 colonies (Section 3.1), so the baseline overstates it by only 0.8%: the population was simulated with the same COLOSS loss rates, and the gap is chance variation in the simulated losses. The prior stratum SDs come from the beta-binomial model of Section 2.4 with the same assumed risk SD of 0.15: each beekeeper has $E(y_i) = p_h m_i$ and $V(y_i) = m_i p_h(1-p_h) + m_i(m_i-1)\,0.15^2$, and S_h^2 is the variance of $E(y_i)$ within the stratum plus the mean of $V(y_i)$. This gives $S_h = 2.92$, 14.40 and 78.37 colonies, against true values of 2.86, 13.72 and 87.13 (Table 2), so even with the generating model, a best case that a real planner would not have, the planner underestimates the commercial SD by about 10%. For the ratio estimator the same model gives residual SDs of $S_{eh} = 2.44$, 13.88 and 67.56, and with full response Neyman allocation, rounded up in each stratum, needs at least $n = 1{,}045$ respondents, the minimum for the primary estimator. The plan keeps the expansion estimator's larger requirement of 1,255 respondents, all 320 commercial, 687 hobby and 248 side-line beekeepers, as a safety margin (Section 3.3.1).

3.2.2 Finite population correction, design effect and non-response (B4)

Without the finite population correction and the take-all rule, Neyman allocation would need $n_{0,\mathrm{st}} = 3{,}198$ respondents; together they cut this to 1,255. For SRSWOR, with the planner's overall SD of 17.65 colonies from the same model, $n_0 = 26{,}883$ exceeds N, and the correction reduces it to 7,289. Beekeepers are sampled individually, not in clusters, so no intracluster correlation enters the design effect: at 1,255 respondents, the expansion estimator has 0.053 times the variance of SRSWOR.

The planned response rate is 77%, the highest of the rates reviewed in Section 2.1.1, reached with personal contact, which the telephone follow-up of Section 2.6 provides here; applying this rate is still an assumption, and non-response is assumed random within strata. The commercial stratum cannot be topped up: about 246 of its 320 beekeepers respond, and the variance left in that stratum, $N_h^2(1 - r_h/N_h)S_h^2/r_h$ with $r_h = 246$ respondents, uses 51% of V_0; below a commercial response rate of 63% (56% for the ratio estimator) it would exceed V_0, so no sample size could reach the target. The hobby and side-line strata make up for it with 1,205 and 436 respondents instead of 687 and 248, for 1,887 respondents in all. Dividing each stratum's respondents by 0.77 and rounding up gives 2,452 beekeepers to issue.

3.2.3 Allocation across strata (B5)

Hobby operations have the largest share of $\sum_h N_h S_h$, so they receive the most respondents, at a sampling fraction of 17%; commercial operations have almost as large a share from only 320 beekeepers, so that stratum is taken in full, and side-line operations are sampled at 86% (Table 3).

Table 3: Neyman allocation with a take-all commercial stratum, for a 95% CI half-width of 5% of the baseline. Respondents n_h are shown under full response and under the planned 77% response rate; the issued sample is $n_h^* = n_h/0.77$, rounded up.

				Respondents n_h			
Stratum	N_h	Prior S_h	Share of $\sum N_h S_h$	Full response	77% response	Issued n_h^*	n_h^*/N_h
HOB	9,020	2.92	43.2%	687	1,205	1,565	17.4%
SID	660	14.40	15.6%	248	436	567	85.9%
COM	320	78.37	41.2%	320	246	320	100.0%
Total	**10,000**		**100.0%**	**1,255**	**1,887**	**2,452**	**24.5%**

3.2.4 Sample size against the margin of error (B6, B7)

Figure 2 plots the respondents needed against the relative margin of error: stratified sampling needs far fewer than SRSWOR at every margin. With 77% response the curve rises steeply as the margin narrows, and no sample size reaches margins below 3.8%, where non-response alone uses up the variance budget even if every beekeeper is issued. Figure 3 repeats the calculation for assumed values of T equal to 25%, 50%, 75% and 100% of the baseline, with full response (solid lines) and 77% response (dashed).

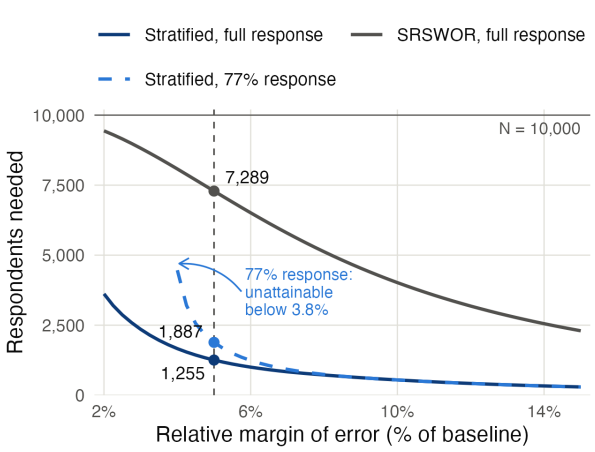


Figure 2: Respondents needed against the relative margin of error. The vertical dashed line marks the 5% target.

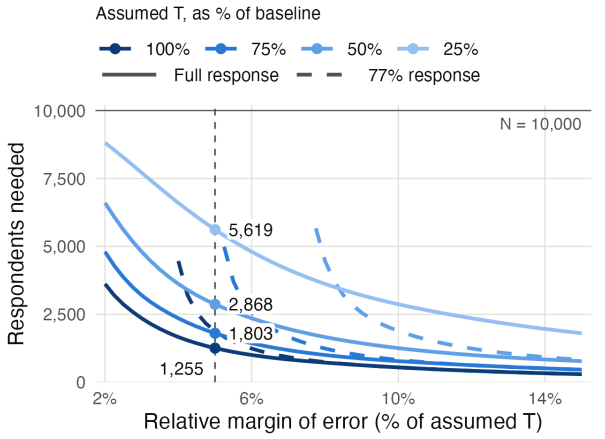


Figure 3: Respondents needed for assumed values of T equal to 25%, 50%, 75% and 100% of the baseline (stratified design), with full and 77% response.

3.2.5 Quantities held fixed and discussion (B8–B10)

Figure 3 holds fixed the population size N and the N_h, the prior S_h, the 95% confidence level, Neyman allocation with the take-all and minimum rules, the finite population correction and, for each set of curves, the response rate. Only the assumed value of T, θT_0 with $\theta = 0.25, 0.5, 0.75, 1$, changes, and with it the target half-width $e\,\theta T_0$. The analysis for proportions (B9) does not apply: for a total the requirement depends on S_h relative to T_0 rather than on $P(1-P)$, so it has no maximum at $P = 0.5$ and instead grows as the assumed total falls.

A smaller assumed total with the same stratum SDs means a smaller absolute half-width, and so a larger sample. At a 5% margin the requirement grows from 1,255 respondents at the baseline to 1,803, 2,868 and 5,619 at 75%, 50% and 25% of it, and at 50% and 25% the side-line stratum is taken in full as well. A lower total would in practice come with smaller SDs, so these figures are conservative. At 77% response the 5% margin is out of reach once T is 75% of the baseline or less, as even issuing every beekeeper would give margins of at least 5.01%, 7.52% and 15.03%. Prior SDs 20% or 30% higher would raise the requirement at 77% response to 3,526 or 6,052 respondents, and the minimum commercial response rate from 63% to 71% or 74%. The design matters as much as the assumed value:

at the baseline, proportional allocation ($n_h \propto N_h$) would need 6,528 respondents and SRSWOR 7,289, in line with $V_{\text{opt}} \leq V_{\text{prop}} \leq V_{\text{SRS}}$. Neyman allocation gains so much because the stratum SDs differ widely: it samples the variable strata heavily and the homogeneous hobby stratum lightly.

3.3 Sample selection (C1–C6)

3.3.1 Final sample size (C1)

The final sample issues 2,452 beekeepers, 24.5% of the register: 1,565 hobby, 567 side-line and all 320 commercial beekeepers (Table 3). This is the Neyman-allocated sample expected to give the expansion estimator a 95% CI half-width of at most 5% of the baseline once the planned 77% response rate is allowed for (Section 3.2.2), and it should yield 1,887 respondents, including about 246 commercial beekeepers. For the ratio estimator, the minimum at 77% response is only 1,324 respondents (Section 3.2.1), so the sample carries a safety margin of 43% more respondents: it meets the bound on the true SE even if the prior residual SDs are 16% too small, and allows for the estimated SE varying between samples (Section 3.4.6).

3.3.2 Selection and realised sample (C3, C5)

`R/03_sample.R` sorts the register by stratum and draws an independent SRSWOR of n_h^* beekeepers in each stratum with `sampling::strata()`, using `set.seed(2026)`; the core of the procedure is listed in Appendix C. Fieldwork is then simulated: each issued beekeeper responds with probability 0.77, independently within strata, and answers are kept only for respondents. The draw gave 1,921 respondents, a response rate of 78.3%, with at least the planned number in every stratum (Table 4).

Table 4: The realised sample: issued sample, inclusion probabilities $\pi_{hi} = n_h^*/N_h$, design weights $d_{hi} = 1/\pi_{hi}$, planned and actual respondents, and the non-response-adjusted weights $w_{hi} = N_h/r_h$.

					Respondents			
Stratum	N_h	Issued n_h^*	π_{hi}	d_{hi}	Planned	Actual r_h	Response rate	$w_{hi} = N_h/r_h$
HOB	9,020	1,565	0.174	5.764	1,205	1,220	78.0%	7.393
SID	660	567	0.859	1.164	436	441	77.8%	1.497
COM	320	320	1.000	1.000	246	260	81.3%	1.231
Total	**10,000**	**2,452**			**1,887**	**1,921**	**78.3%**	

Every beekeeper on the register belongs to exactly one stratum and has a known, positive inclusion probability: 0.174 in the hobby stratum, 0.859 in the side-line stratum and 1 in the commercial stratum. Over the register the probabilities sum to 2,452, the issued sample size, and the code checks that the issued sample's design weights and the respondents' adjusted weights each sum to $N = 10{,}000$.

3.3.3 Data files (C2, C4, C6)

The population and the sample are saved as CSV files, and `data/population.csv`, generated by `R/01_population.R` (Section 2.4), holds one row for each of the 10,000 beekeepers with a beekeeper identifier, the stratum and operation-size class, migratory practice, colonies going into winter m_i, colonies lost y_i and the loss percentage. `data/sample.csv` holds one row for each of the 2,452 issued beekeepers, with the identifier, the stratum and operation-size class, the register's colonies m_i (an auxiliary variable), N_h and n_h^*, the inclusion probability π_{hi}, the design weight d_{hi}, a response flag, r_h, the adjusted weight w_{hi}, migratory practice and the outcome y_i. The last three are filled in for the 1,921 respondents and left blank for the 531 non-respondents, whose answers were not obtained.

3.4 Estimation (D2–D8)

3.4.1 Estimate of the total (D2, D3)

The ratio estimator gives an estimated total of $\hat{T}_R = 40{,}090$ colonies lost, a loss rate of $\hat{T}_R/M = 14.8\%$, with estimated variance 692,299, SE 832 and 95% CI from 38,459 to 41,720 colonies (Table 5). The coefficient of variation (CV) is 2.1%, and the design effect, the ratio of this variance to that of SRSWOR with the same number of respondents, is 0.055. Hobby, side-line and commercial operations

account for 15,285, 7,797 and 17,008 colonies, or 38%, 19% and 42% of the losses. The expansion estimator gives $\hat{T} = 39{,}762$ with SE 937 and design effect 0.069 (0.053 was expected at the planning stage, Section 3.2.2), so the register's colony counts cut the SE by 11%.

Table 5: Estimates of the total colonies lost from the realised sample, with the expansion estimator and the ratio estimator (bold, the primary estimator), and from the SRSWOR benchmark, against the true total. The error is relative to T.

Estimator	Estimate	Error	SE	95% CI	CV	Deff	Respondents
Naive, $N\bar{y}$	109,271	+161.1%					1,921
Design weights d_{hi} only	31,542	−24.6%					1,921
Expansion $\hat{T}$, weights w_{hi}	39,762	−5.0%	937	37,926–41,598	2.4%	0.069	1,921
Ratio estimator $\hat{T}_R$	**40,090**	**−4.2%**	**832**	**38,459–41,720**	**2.1%**	**0.055**	**1,921**
SRSWOR benchmark	40,541	−3.1%	2,797	35,059–46,024	6.9%	1.000	1,903
True total T	41,853						

3.4.2 Inclusion probabilities and weights (D4)

Each beekeeper's inclusion probability is $\pi_{hi} = n^*_h/N_h$, with the values in Table 4. Its inverse, the design weight d_{hi}, is the number of register beekeepers that each issued beekeeper represents: 5.76 hobby, 1.16 side-line or 1 commercial beekeeper. Non-response leaves r_h respondents in each stratum, so the weights are multiplied by n^*_h/r_h, giving $w_{hi} = N_h/r_h = 7.39$, 1.50 and 1.23; even a commercial respondent, selected with certainty, stands in for the commercial beekeepers who did not respond. The ratio estimator multiplies these weights by $M_h/\hat{X}_h$, where $\hat{X}_h = \sum_i w_{hi} m_i$ is the respondents' weighted colony count, so that it matches the register's M_h; the factors are 1.015, 1.008 and 1.002. Within each stratum the adjusted weights sum to N_h, so the respondents represent the whole register.

3.4.3 Naive, design-adjusted and true values (D5)

Ignoring the design, N times the unweighted mean of the respondents gives 109,271 colonies, 161% too high: commercial operations are 13.5% of the respondents but only 3.2% of the register. The design weights alone give 31,542, 24.6% too low, because the weight of the 531 non-respondents is lost; the adjusted weights pass that weight to the respondents in the same stratum and give an estimate 5.0% below the true total of 41,853, and the ratio estimator one 4.2% below.

3.4.4 SRSWOR benchmark (D6)

An SRSWOR of 2,452 beekeepers, the same issued size, drawn with `set.seed(2027)` and subject to the same 77% response, gave 1,903 respondents and an estimate of 40,541 with SE 2,797 (Table 5). With almost the same number of respondents, its variance is 11.3 times that of the ratio estimator and its CI is 3.4 times as wide.

3.4.5 Empirical and estimated standard errors (D7)

Over the 10,000 repetitions of Section 3.5, the ratio estimator has an empirical SE of 972 against an average estimated SE of 969, so its variance estimator is accurate, as is the expansion estimator's (empirical 1,123, average estimated 1,124); for SRSWOR the average estimated SE of 3,761 is 4% below the empirical 3,908. The expansion estimator's realised SE of 937 is below its theoretical value of 1,046, computed from the true S_h for the realised r_h: the sample variances came out low in all three strata.

3.4.6 Precision bound and the effect of the design (D8)

The ratio estimator's realised half-width is 1,631 colonies, 3.9% of the baseline, so this sample meets the 5% bound, as it does with the expansion estimator (4.4%). Both CIs nevertheless miss the true total, as a 95% CI should in about one sample in 20: the respondents' losses came out low in all three strata, the stratum means by 1.0 to 1.5 SEs (computed from the true S_h). Over repeated samples the ratio estimator meets the bound in 97% of samples and the expansion estimator in only 27%, with an average half-width of 5.2% of the baseline. The formula of Section 2.8 bounds the true SE, a bound the estimated half-width meets about half the time even under a correct plan: a plan with the true SDs

(2,187 respondents) meets it with the expansion estimator in 51% of samples (`R/05_simulation.R`), and the lower 27% reflects the prior, which underestimated the commercial SD (Section 3.2.1). Meeting the bound in 80% or 90% of samples thus needs a margin, which for the expansion estimator would mean issuing about 3,800 or 4,300 beekeepers rather than 2,452; the ratio estimator gets it from the planned sample (Section 3.3.1). The design is what makes the bound reachable at all: stratification and the colony counts cut the variance to 0.055 times that of SRSWOR, which never meets the bound with this many respondents.

3.5 Simulation study (E3–E5)

3.5.1 Performance measures (E3, E4)

Against the true total $T = 41{,}853$ (Table 6), all three estimators are essentially unbiased: the biases of -3 colonies for the ratio estimator, -9 for the expansion estimator and -56 for SRSWOR are 0.3, 0.8 and 1.4 Monte Carlo SEs from zero. With $R = 10{,}000$, the ratio estimator's CIs cover T in 94.8% of repetitions (Monte Carlo SE 0.2 percentage points), within Monte Carlo error of the nominal 95% (1.0 SE below). The expansion estimator's cover it in 94.5%, slightly but significantly below 95% (2.2 Monte Carlo SEs), and SRSWOR's in only 91.4% (SE 0.3). The expansion estimator and SRSWOR miss mostly on the low side, in 419 of 550 and 818 of 859 misses. Samples that happen to contain fewer of the large commercial operations give low estimates together with low estimated SEs (correlations 0.61 and 0.81), so their CIs are too narrow on the low side; the take-all stratum removes most of this effect, the rest coming mainly from its non-respondents. The ratio estimator weakens the link further (correlation 0.50) and its misses are less lopsided (369 of 522 low), which brings its coverage closer to 95%.

Table 6: Performance of the stratified design, with the ratio and expansion estimators, and of SRSWOR over 10,000 repetitions, against the true total $T = 41,853$. Monte Carlo SEs of the bias and of the coverage (in percentage points) are in brackets. The last row repeats the simulation (2,000 repetitions each) with half / twice the response odds for a beekeeper who lost half their colonies (Section 3.5.2).

	Stratified (Neyman)		
Measure	Ratio $\hat{T}_R$	Expansion $\hat{T}$	SRSWOR
Bias (Monte Carlo SE)	−3 (10)	−9 (11)	−56 (39)
Relative bias	−0.01%	−0.02%	−0.13%
Empirical SE	972	1,123	3,908
Average estimated SE	969	1,124	3,761
RMSE	972	1,123	3,908
Coverage of the 95% CI (Monte Carlo SE)	94.8% (0.2)	94.5% (0.2)	91.4% (0.3)
Average CI width	3,800	4,407	14,743
Design effect (variance ratio)	0.062	0.083	1.000
Samples meeting the 5% bound	96.6%	26.8%	0.0%
Average respondents	1,888	1,888	1,888
Relative bias if response depends on losses	−6.2% / +5.1%	−6.2% / +5.2%	−5.7% / +4.6%

3.5.2 Comparison with SRSWOR (E5)

With the same average of 1,888 respondents, the stratified design with the ratio estimator has 0.062 times the empirical variance of SRSWOR (the empirical design effect), an RMSE of 972 against 3,908, and CIs about a quarter as wide on average (3,800 against 14,743 colonies), with better coverage. Its variance is 25% below that of the expansion estimator, which has 0.083 times the variance of SRSWOR. The SRSWOR estimator uses a single adjustment N/r; post-stratifying it by size class would give about the precision of proportional allocation, which needs 6,528 respondents against 7,289 (Section 3.2.5), so the stratified design gains mainly through Neyman allocation and the take-all stratum, not through weighting within strata. Figure 4 shows how tightly the ratio estimator clusters around T compared with SRSWOR. These results assume that non-response is random within strata. If response depends on the losses, which Gray et al. (2020) report can happen in either direction, the weighting cannot remove the bias: when a beekeeper who lost half their colonies has half or twice the response odds of one who lost none, the ratio estimator is off by −6.2% or +5.1% and its CIs cover T in only 23% or 41% of samples, against 79% or 94% for SRSWOR, whose wider CIs absorb more of the bias (Table 6).

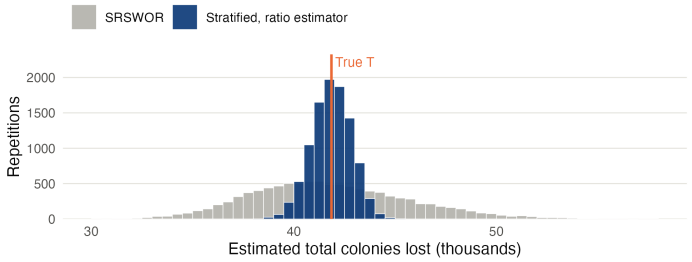


Figure 4: Sampling distributions of the estimated total over 10,000 repetitions for the stratified design with the ratio estimator and for SRSWOR, with the true total marked.

4 Conclusion and Recommendations

4.1 Summary and conclusion

In contrast to the surveys reviewed in Section 2.1, the design developed here gives every registered beekeeper a known, positive inclusion probability, so its estimates carry design-based SEs. The survey estimates that registered beekeepers lost 40,090 colonies over winter 2025/26 (95% CI 38,459 to 41,720), 14.8% of the colonies going into winter. Commercial operations, only 3.2% of beekeepers, account for 42% of the estimated losses, hobby operations for 38% and side-line operations for 19%. The CI narrowly misses the true total of 41,853, as a 95% CI does in about one sample in 20.

Allowing for 77% response, the survey needs 1,887 respondents from 2,452 issued beekeepers: 1,565 hobby, 567 side-line and all 320 commercial beekeepers, planned with the expansion estimator's larger variance as a safety margin for the ratio estimator. Stratification by operation size with Neyman allocation is far more efficient than SRSWOR: at full response it needs 1,255 respondents against 7,289 for the same precision, and over 10,000 repetitions the ratio estimator's variance is 0.062 times that of SRSWOR. The ratio estimator meets the 5% bound in 97% of samples, and its CIs cover the true total in 94.8%, against 91.4% for SRSWOR. Without the colony counts, the expansion estimator meets the bound in only 27% of samples, because the prior understated the commercial SD.

4.2 Limitations

The results apply to the registered beekeepers of a simulated population: the distributions of colonies within the size classes, the loss-risk SD of 0.15 and the independence of migration and operation size are assumptions, and the calibration pools 36 countries, so regional differences are not represented (Section 2.4). Fieldwork is simulated with response at random within strata, so the weighting removes non-response bias by construction; if response depends on the losses, the estimates are off by about 5–6% and their CIs undercover badly (Section 3.5.2). The plan also depends on prior SDs, here from the generating model itself, as the COLOSS reports cited give no between-operation SD of loss risk; they understated the commercial SD (Section 3.4.6), and understating all of them by 20% would almost double the sample needed (Section 3.2.5).

4.3 Recommendations for further work

A real survey should keep the ratio estimator and a safety margin in the plan, as here; the expansion estimator alone would need about 4,300 issued beekeepers to meet the bound in 90% of samples, and finer strata on colony counts (Section 2.5) would cut the sample further. The response rate in the take-all commercial stratum is the main risk: below 63% on the planning values no sample size reaches the target, so intensive follow-up of the 320 commercial operations matters more than a larger sample of hobby beekeepers. A short non-response follow-up asking only for colonies lost would test the MAR assumption, on which the estimates depend (Section 3.5.2).

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Appendix A — Calibration of the simulated population

Table A1: Composition of the simulated population by operation size, compared with the COLOSS survey of winter 2019/20 (Gray et al., 2023). The COLOSS mean colonies and shares of colonies are computed from the numbers of beekeepers and colonies in each class reported there. The stratum weights W_h equal the COLOSS shares of beekeepers (90.2%, 6.6% and 3.2%). In the COLOSS survey 81.7% of beekeepers did not migrate their colonies, so the simulated migration probability of 0.183 slightly overstates migration: the remaining 18.3% includes 163 beekeepers who answered that they did not know.

			Mean colonies			Share of colonies		
Stratum	N_h	W_h	Sim.	COLOSS	Colonies M_h	Sim.	COLOSS	Migratory
Hobby (HOB), up to 50 colonies	9,020	0.902	10.8	10.8	97,216	35.9%	35.5%	18.8%
Side-line (SID), 51–150 colonies	660	0.066	86.5	88.2	57,062	21.1%	21.2%	20.0%
Commercial (COM), more than 150 colonies	320	0.032	364.7	371.8	116,709	43.1%	43.3%	16.9%
Total	**10,000**	**1.000**	**27.1**	**27.5**	**270,987**	**100.0%**	**100.0%**	**18.8%**

Appendix B — Sources of non-sampling error

Table B1: Main sources of non-sampling error.

Source	Cause	Likely effect on $\hat{T}$	Mitigation
Coverage	Duplicate entries; registered beekeepers who no longer keep bees; outdated contact details or colony counts	Duplicates inflate some selection chances; ineligible beekeepers reduce the effective sample; uncontactable ones are equivalent to non-respondents; wrong strata lower efficiency	Removal of duplicates; eligibility question; colony count asked again
Non-response	Selected beekeepers who cannot be reached, decline or do not reply	Bias in either direction if non-respondents' losses differ systematically from respondents'	Reminders; CATI follow-up; incentives; weighting within strata (Section 2.6)
Response	Recall error in colony counts; ambiguous cases such as weak, merged or split colonies	Measurement error in y_i and m_i	Fixed reference dates; COLOSS case definitions; trained interviewers; pretest
Processing	Data-entry errors, such as more colonies lost than held	Wrong values for a few beekeepers	Range checks ($0 \le y_i \le m_i$); checks against the register

Appendix C — Selection and estimation code

The core of the selection procedure in `R/03_sample.R` (Section 3.3.2) draws an independent SRSWOR in each stratum:

```
set.seed(SEED_SAMPLE)                                    # 2026
frame <- pop[order(match(pop$stratum, H), pop$bk_id), ]  # sorted by stratum
s <- sampling::strata(frame, stratanames = "stratum",
                      size = as.numeric(n_iss),          # 1,565/567/320
                      method = "srswor")
smp <- sampling::getdata(frame, s)                       # adds Prob
```

The survey design object in `R/04_estimation.R` (Section 2.9) declares the strata, the adjusted weights and the finite population correction, and gives both estimators, the ratio estimator with the register's colony totals M_h:

```
resp <- subset(smp, responded == 1)
des  <- svydesign(ids = ~1, strata = ~stratum, fpc = ~N_h,
                  weights = ~weight, data = resp)
est  <- svytotal(~colonies_lost, des, deff = TRUE)       # expansion estimator
rs   <- svyratio(~colonies_lost, ~colonies_oct, des, separate = TRUE)
est_r <- predict(rs, total = M_h[names(rs$ratios)])      # ratio estimator
```